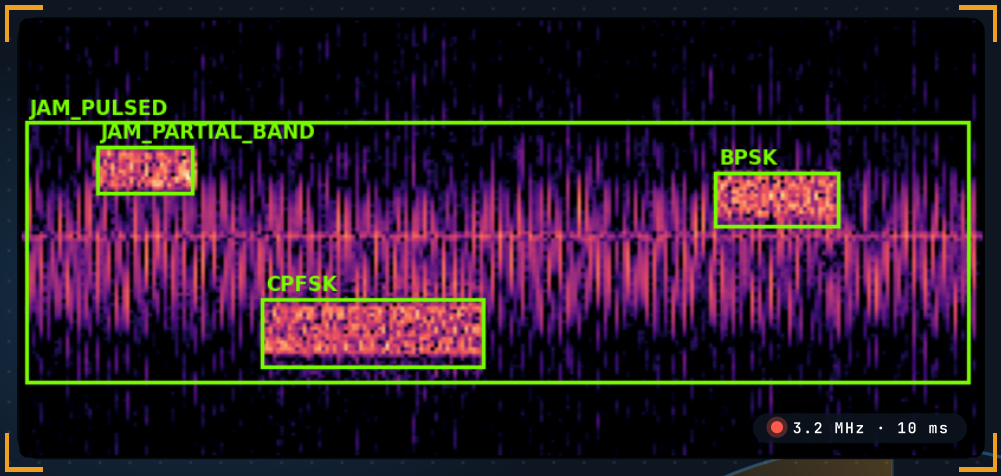
NEXA

SEDIC 26 Strategic Electronic Defence Innovation Challenge 2026



NEXA's own boxes and answers on a simulated capture at 20 dB, not retouched: a pulsed jammer laid over three other signals, and every one still found and named.

TRACK I · RF SIGNAL INTELLIGENCE

Every emitter on the band, named.

NEXA watches a whole band of radio spectrum, finds every transmission in it and tells the operator what each one is: jammer, radar, voice or data. It runs offline on one workstation, and people stay in control.

HOW IT WORKS

1 **Capture** Receiver, IQ file or simulator	2 **Detect** A box on every emitter	3 **Classify** 60 classes, per window	4 **Track** Votes over time	5 **Resolve** Settles look-alikes	6 **Listen** Voice and Morse to text

60
signal classes in 12 families

90%
right signal family, 15 dB, 8-window vote

79.4%
exact class out of 60, SNR ≥ 10 dB, 16-window vote

0
data leaves the machine

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Tri-Branch Deep Fusion Network

for Multi-Class RF Signal Classification

160 µs one window of the signal → raw IQ *CNN* · spectrogram *CNN* · spectrum *CNN* · physics + expert *features* → **Fuse** attention + dense layers → **60 yes / no** two can be true at once → **Track vote** answer + confidence

Three neural networks read each window, fused with measured physics. Look-alikes (QAM / APSK, sizes) are named as a group; every answer has a band: *HIGH* *MEDIUM* *UNCERTAIN*

Named exactly, out of 60.



per emitter

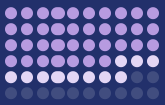
all 60 classes · SNR ≥ 10 dB · 16-window vote · 120,000 unseen test windows. One window: 62.4%



signal family

15 dB · vote over 8 windows · 32 fresh tracks per class

48 / 60



classes named exactly

≥ 50% of the time, 37 of them ≥ 90% · 15 dB · 8-window vote

Right family, family by family · 15 dB · 8-window vote

ASK	100%	FSK	100%	Radar	95%	QAM	94%
Jammers	93%	OFDM / LTE / NR	93%	PSK	91%	Analog voice	86%
Spread spectrum	82%	Pulse comms	70%	APSK	69%	Noise floor	66%

Built for contested spectrum

Sees the attack
Ten jammer types and eight radar waveforms, named next to the signal they sit on.

Says how sure it is
Uncertain answers go to a person, not into the log as fact.

Stays in the room
Detection, speech recognition and AI reading run on one offline machine.

It learns, but only with approval

1. **Operator corrects** — One click.
2. **Analyst approves** — A second person.
3. **Retrain** — Started by a person.
4. **Gate** — Goes live only if better.

NEXA

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Try it

Brochure